Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	Mickey Mouse Clubhouse
Selected Domain	Domain 2
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	Orbyte Collective
Name (Leader)	Leow Wei Xin
Email & Phone (Leader)	018-3907912 (leowweixin@gmail.com)
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small Malaysian businesses face a recurring purchasing challenge: they must restock essential supplies while managing scattered supplier messages, paper invoices and limited buying power. Our team focuses initially on independent cafés and small food and beverage businesses, with similar challenges affecting workshops, salons and neighbourhood shops.

Owners often handle purchasing alongside daily operations. Each restocking cycle requires checking stock, searching for previous prices, requesting quotations and comparing product quantities, delivery charges and payment terms. Without organised records or time to verify every invoice, price increases, incorrect charges and unexpected fees can go unnoticed. Delayed ordering can also cause stock shortages and emergency purchases.

These businesses typically buy quantities too small to qualify for wholesale prices or free delivery. Nearby shops may need identical supplies, but coordinating their orders manually requires time and trust. Buying extra stock individually to reach supplier thresholds ties up cash and risks waste.

This matters economically because avoidable purchasing costs reduce already limited margins and leave less money for wages, operations and business growth. Socially, the burden falls heavily on owner-operated enterprises with limited administrative support, reinforcing the purchasing advantage enjoyed by larger businesses. Overbuying and fragmented deliveries can also create unnecessary waste and transport demand.

Our challenge is to help small businesses reduce this repeated administrative burden and access fairer purchasing opportunities while preserving their independence, cash flexibility and control over every order.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Orbyte Collective is an **AI-powered purchasing and restocking application** for small Malaysian businesses, starting with independent cafés. Designed to integrate with the Orbyte hub, it turns supplier messages, invoice photos and stock updates into a workflow that helps owners compare costs, avoid shortages and access collective buying opportunities.

The application uses three specialised AI agents. An **Intake Agent** extracts purchase and stock information from photos, PDFs and messages. A **Spec-Matching Agent** identifies equivalent products across supplier names, codes and packaging descriptions. A **Supplier Liaison Agent** converts supplier replies into structured quotations and drafts quotation requests or invoice queries for owner approval.

A rules-based engine calculates total delivered costs, suggests reorder timing, checks invoices against approved quotations and identifies compatible orders from nearby businesses. Pooled purchasing is offered only when it reduces each participant's individual cost, enabling bulk pricing while businesses buy only what they need.

A learning layer makes each restocking cycle inform the next. It is essential because changing stock usage and supplier charges can make fixed assumptions unreliable, causing late reorders and misleading comparisons. It analyses confirmed purchasing and stock records to identify usage patterns, recurring delivery charges and quotation discrepancies. It proposes evidence-backed rules, including updated consumption rates and expected delivery fees, for human acceptance. Accepted rules improve future estimates, while invoice checks continue to flag unapproved charges.

The workflow responds automatically to incoming documents, reorder dates and pooled-order changes. Owners confirm records and approve purchases within defined limits.

We apply SDC Phase 1 skills in **GenAI API integration, database architecture and cloud hosting**, using multimodal GenAI APIs, Next.js, TypeScript, Python FastAPI and Supabase PostgreSQL. We use sponsor-provided Replit access to build the prototype, with simulated WhatsApp conversations and Orbyte hub services demonstrating the workflow.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

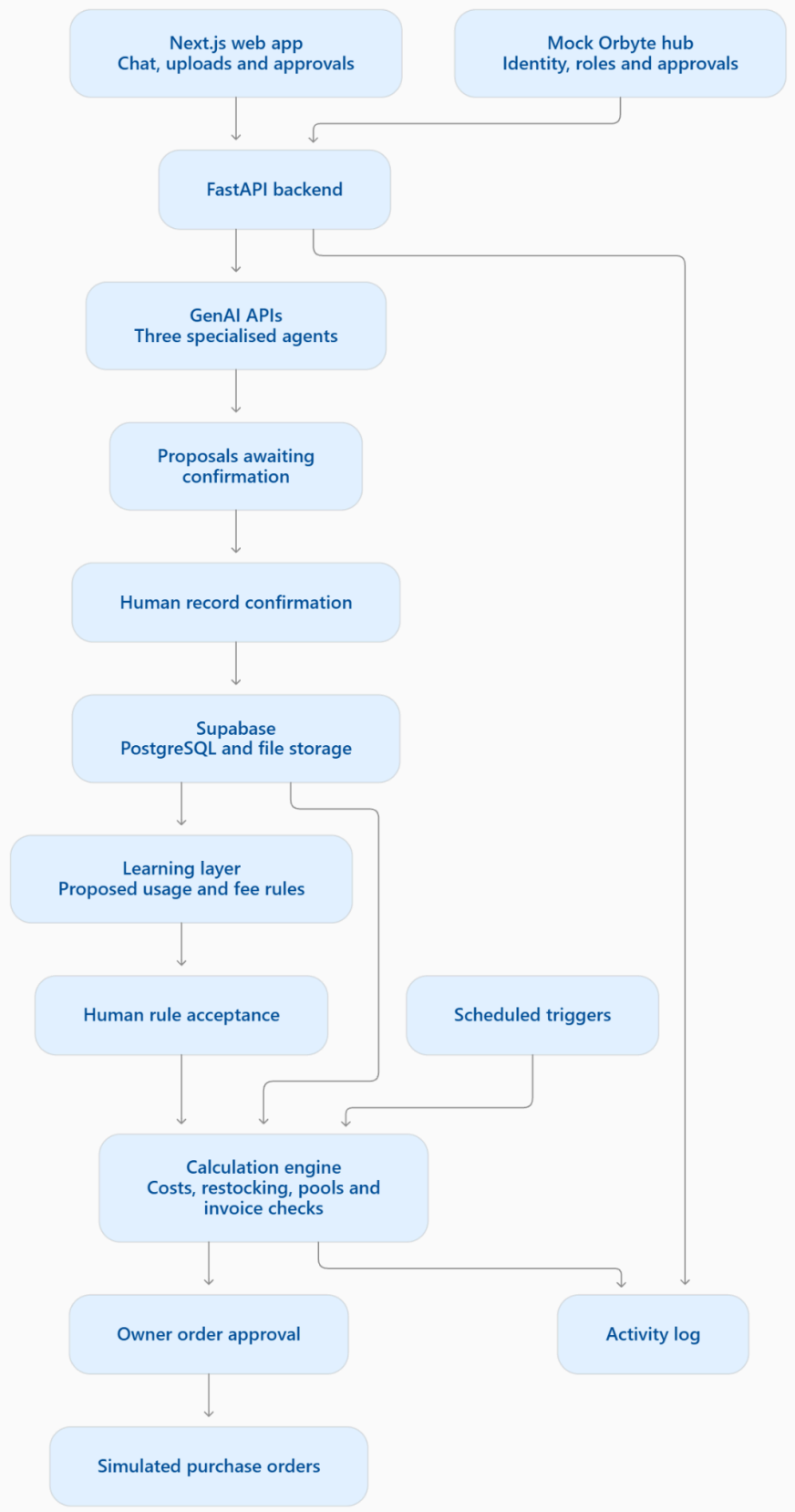
We plan to build and deploy Orbyte Collective using the following technologies:

Component	**Planned Technologies**	**Purpose**
Frontend	Next.js, TypeScript, Tailwind CSS and shadcn/ui	Mobile-friendly interface for document uploads, proposals, approvals and purchasing records
Backend	Python FastAPI and Pydantic	API endpoints, structured data validation and purchasing workflow coordination
Generative AI	Multimodal and text GenAI APIs; provider to be finalised	Extract invoice details, interpret supplier quotations, match product descriptions and draft messages
Product Matching	Multilingual embeddings, with BGE-M3 as a candidate, plus unit-normalisation rules	Compare supplier descriptions against standard product specifications
Database and File Storage	Supabase PostgreSQL and Supabase Storage	Store business records, quotations, approvals, activity logs and uploaded documents
Access Control	PostgreSQL Row-Level Security	Separate each business's data and enforce access permissions
Automatic Triggers	APScheduler or Supabase Cron	Check reorder windows, approval expiry and changes requiring recalculation
Deployment	Vercel for the frontend; Render or Fly.io for the backend; Supabase for managed data services	Host a functional, accessible prototype
Testing	pytest	Verify cost calculations, pooling rules and approval validity

The prototype will simulate WhatsApp conversations and connect to a mock Orbyte hub API for identity, roles and approvals.

The architecture connects workshop elements as follows: the web interface sends documents and messages to the backend; GenAI APIs produce structured proposals; users confirm these before database storage; the calculation engine uses confirmed records and accepted learning rules to generate recommendations. Owners approve orders before simulated purchase orders are produced. Activity is recorded throughout.

Official SDC sponsor-provided services: We use Replit to build and run our prototype, using the free sponsor-provided access offered through the SDC organisers.


Next.js web app
Chat, uploads and approvals
Mock Orbyte hub
Identity, roles and approvals
FastAPI backend
GenAI APIs
Three specialised agents
Proposals awaiting confirmation
Human record confirmation
Supabase
PostgreSQL and file storage
Learning layer
Proposed usage and fee rules
Human rule acceptance
Scheduled triggers
Calculation engine
Costs, restocking, pools and invoice checks
Owner order approval
Activity log
Simulated purchase orders

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design-thinking approach centres on the recurring restocking tasks of small business owners.

Empathise: We used the proposal's desk research and owner scenarios to understand purchasing through WhatsApp, phone calls and paper invoices. Planned interviews with 3–5 owners will validate their workload, invoice discrepancies and willingness to pool orders.

Define: We narrowed the challenge to one recurring burden: restocking without clear price records, efficient comparisons or sufficient purchasing power.

Ideate: We developed two complementary modes: individual purchasing support that delivers immediate value, and optional pooled purchasing when each participating business benefits.

Prototype: Our proposed mobile-friendly interface uses familiar chat interactions, simple approval cards, source evidence and multilingual support. Three AI agents interpret messy inputs, while explicit code handles costs and approval rules.

Test and prepare: Before Phase 2, we plan owner usability walkthroughs, supplier validation of split invoicing and delivery, and extraction tests using 30 synthetic Malaysian invoices. Automated tests will verify pooling calculations and approval changes. Cloud hosting, business-level data isolation and activity logs support deployment preparation.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary: SDG Target 8.2** — Increase economic productivity through technological upgrading and innovation. Orbyte Collective reduces repetitive purchasing administration through AI-assisted document processing and automated comparisons. **Supporting: SDG Target 8.3** — Support entrepreneurship and MSME growth. The application helps small businesses manage purchasing costs and access collective buying opportunities. These are project-level contributions, rather than direct measurements of the UN's national indicators. [1]
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Published context:** A 2025 study involving 100 respondents responsible for purchasing in Malaysian small restaurants reported that **37% used frequent supplier comparisons** as a purchasing strategy, indicating room for more systematic comparison. This is a study-specific finding, not a national prevalence estimate. [2] **Provisional operational baseline:** We estimate **120 minutes of active owner/staff time per business per weekly restocking cycle**, covering record entry, quotation requests, comparisons, order preparation and invoice checking. This is a team planning assumption, not a published finding. Before prototype testing, we will

Metric Category	Your Response
	validate or replace it through timed manual walkthroughs with 3–5 target users.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary: Percentage reduction in active restocking administration time.** Formula: **(manual time − prototype-assisted time) ÷ manual time × 100**. Timing includes document corrections and human approvals, but excludes passive waiting for supplier replies. **Secondary: Percentage reduction in total delivered purchasing cost**, comparing the same specifications and quantities against an available individual-order quotation, including delivery charges and any platform fees.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**Achieve at least a 30% reduction in active administration time** across at least 10 paired manual-versus-prototype scenarios involving 3–5 target users. Against the provisional baseline, this means reducing **120 minutes to 84 minutes or less** per equivalent cycle. Both methods must complete the same tasks accurately. **Secondary target:** Demonstrate **at least 5% lower delivered cost for every participating business in at least three eligible pooled-order scenarios**, without increasing required quantities. Supplier quotations should support these comparisons; synthetic scenarios will be labelled as simulations. These are targets to test, not achieved results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

( ) Yes (/) No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

( ) Yes (/) No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

( ) Yes (/) No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Leow Wei Xin

25th October 2026